

Assignment 1: Opinion Network Formation

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1. Team Name

Team Name: [TEAM NAME TBD]. Replace this placeholder with the official group name before submission. Member names appear in Section 7.

2. GitHub Link to Code

Repository: [GITHUB URL TBD]. The repository should contain the `src/` pipeline, `data/Survey\_Results\_UC.csv`, `requirements.txt`, and this report. After you push the local project, paste the public URL here and confirm the repository is accessible.

3. Dataset Documentation

The class survey asked 60 Likert-scale statements covering four themes: Technology (T01-T15), Education (E01-E15), Ethics/Society (S01-S15), and Environment (V01-V15). After loading the CSV with UTF-8-SIG (to strip the BOM on the id column), we obtained 91 response rows and 60 items. 5 rows had fewer than 10 valid Likert answers (entirely blank submissions) and were dropped, leaving 91 nodes. Each remaining response is an ordinal opinion, not a social contact, so the network we construct is a derived similarity graph rather than a reported friendship graph.

We mapped Strongly Disagree=1, Disagree=2, Neutral=3, Agree=4, Strongly Agree=5. Non-scale tokens such as 'No Comments' were treated as missing (NaN) and excluded from pairwise comparisons. Missing cells: 242 (4.43% of all answers). Most missing cells are blank survey fields rather than a second scale; they are excluded from pairwise comparisons. The remaining rate is low enough that Pearson correlation on pairwise-complete coordinates remains stable.

We interpret each respondent as a 60-dimensional opinion vector. Category prefixes are used both to document the instrument and to build four theme-specific sub-networks. Higher scores mean stronger agreement with the (generally pro-technology, pro-learning, pro-ethics, pro-environment) wording of the items. Because almost every statement is positively framed, a high mean is a 'progressive / high-agreement' profile rather than a mix of opposing ideologies. Missingness is spread across all four blocks rather than concentrated in a single theme, so category sub-networks remain comparable.

4. Pipeline Followed

Nodes are respondents (students). Similarity is the Pearson correlation of their Likert vectors (cosine similarity after mean-centering). Pearson is used instead of raw cosine because nearly all answers sit on the Agree side of the scale, which would otherwise make every pair look almost identical. An undirected edge exists when either student is among the other's 8 nearest neighbors and the correlation is positive. Edge weight is the correlation. kNN is preferred over a complete weighted graph because a dense 96-node clique has trivial density and no community structure worth reporting.

Step-by-step: (1) encode Likert answers to 1-5; (2) compute a 96 x 96 Pearson similarity matrix using only coordinates observed for both people; (3) for each respondent keep the 8 most similar others and add an undirected edge if the relation holds in either direction and the correlation is positive; (4) repeat the same construction independently on the 15 Technology, Education, Ethics, and Environment items; (5) compute global graph statistics, degree/betweenness/eigenvector centrality, and Louvain communities (weight = similarity, resolution = 1); (6) characterize each community by its mean Likert score in the four themes.

The resulting overall network has 91 nodes, 357 edges, density 0.087, average degree 7.85, and 1 connected component(s) (largest component size 91). Louvain is run on this weighted graph; category networks are analyzed separately so we can see whether Technology opinions fragment the class more than Environment opinions.

5. Analysis and Visualizations

Global metrics (overall network): density 0.087, average clustering 0.168, average shortest path in the giant component 2.59, diameter 6. Louvain found 5 communities with modularity 0.369. Community sizes: {"0": 11, "1": 23, "2": 15, "3": 20, "4": 22}. A modularity well above 0 indicates that similarity is not uniform: some groups of students agree with each other more than with the rest of the class.

Table 1. Global metrics of the overall opinion network

Nodes	Edges	Density	Avg deg	Clustering	ASP	Diameter	Modularity
91	357	0.087	7.85	0.168	2.59	6	0.369

Eigenvector centrality ranks respondents who are similar to other well-connected (high-similarity) students - a 'consensus core'. Top eigenvector ids: 114 (0.307), 104 (0.298), 81 (0.259), 61 (0.221), 110 (0.217). Betweenness highlights bridges between communities. Top betweenness ids: 104 (0.166), 114 (0.143), 95 (0.089), 90 (0.080), 77 (0.076). High-betweenness students are not necessarily the most typical; they sit on short paths linking otherwise distinct opinion clusters.

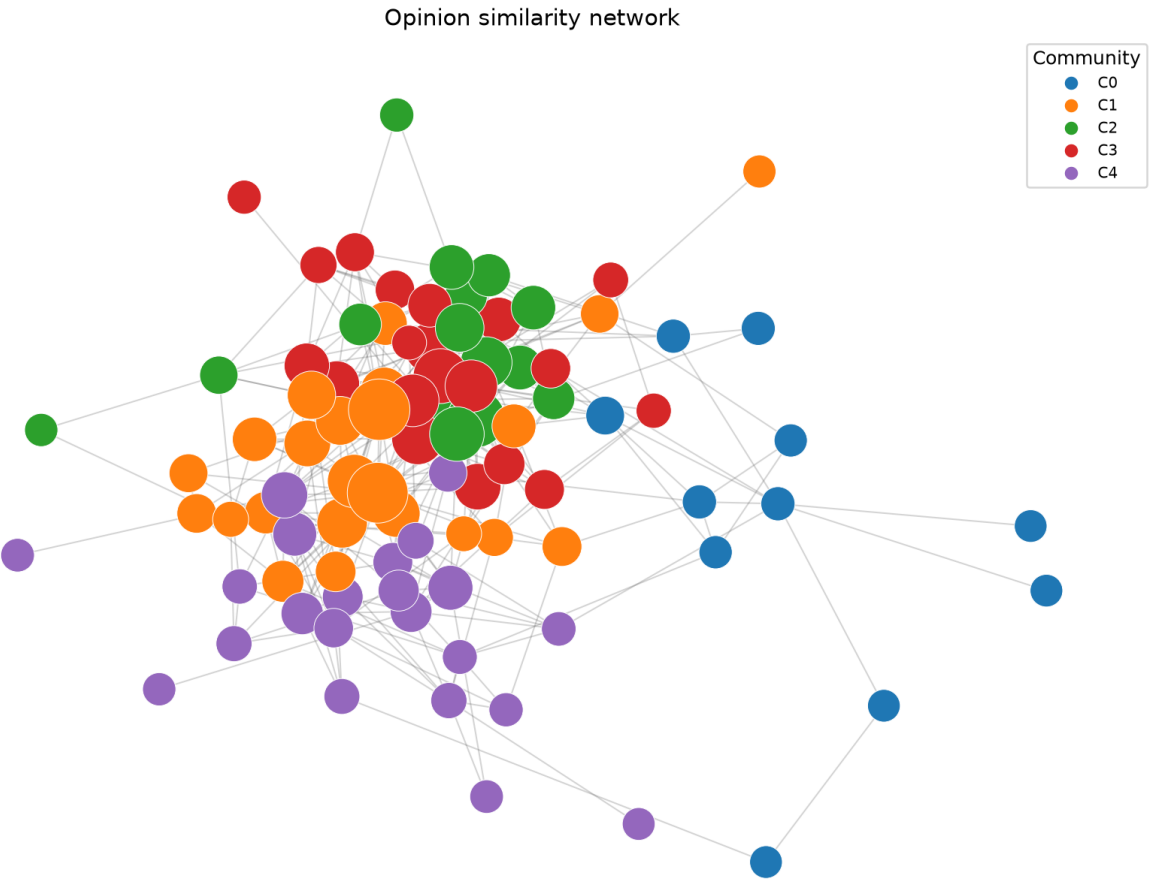


Figure 1. Overall mutual-kNN opinion network. Node color is Louvain community; node size is eigenvector centrality. Tight color blobs are groups of students whose 60 answers are close in cosine space.

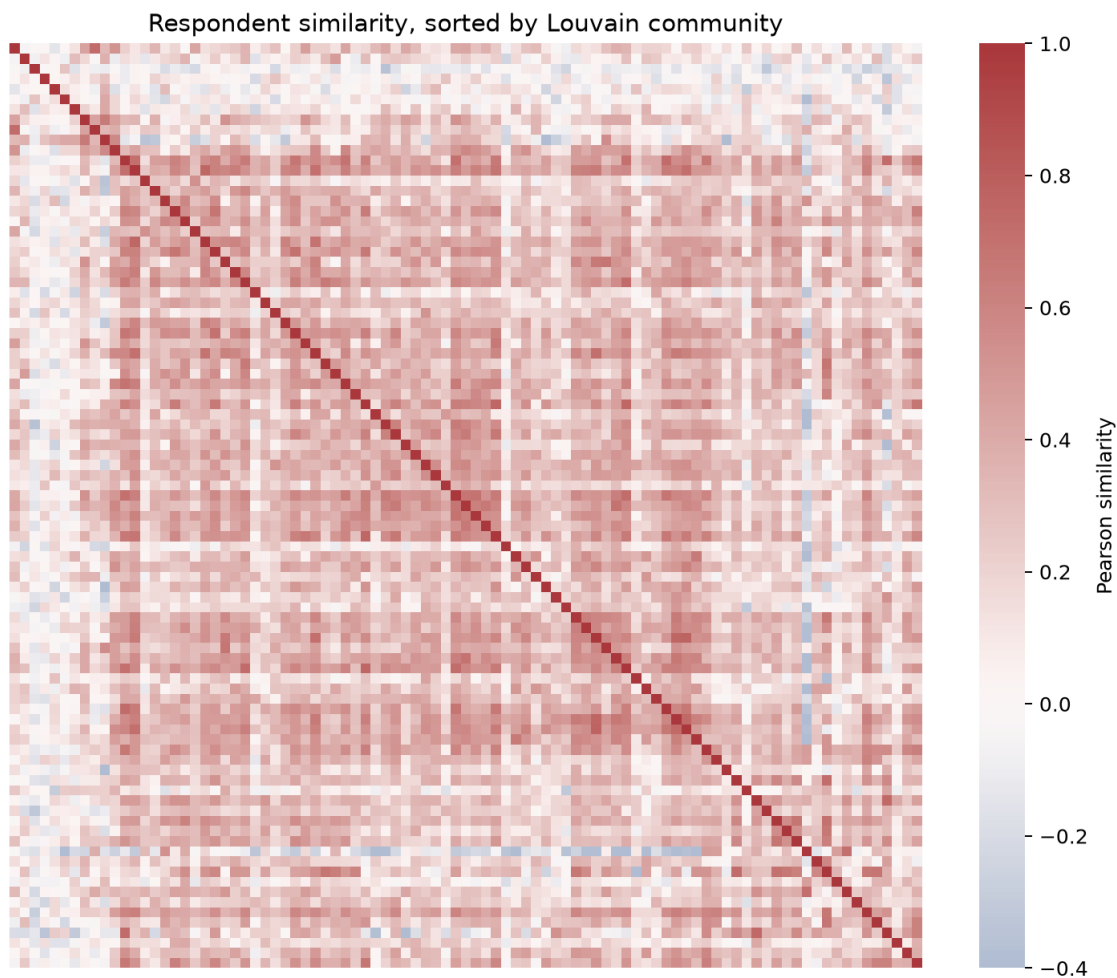


Figure 2. Pearson similarity heatmap with respondents sorted by community. Warm blocks on the diagonal show within-community agreement; cooler off-diagonal cells show weaker or opposing profile shapes.

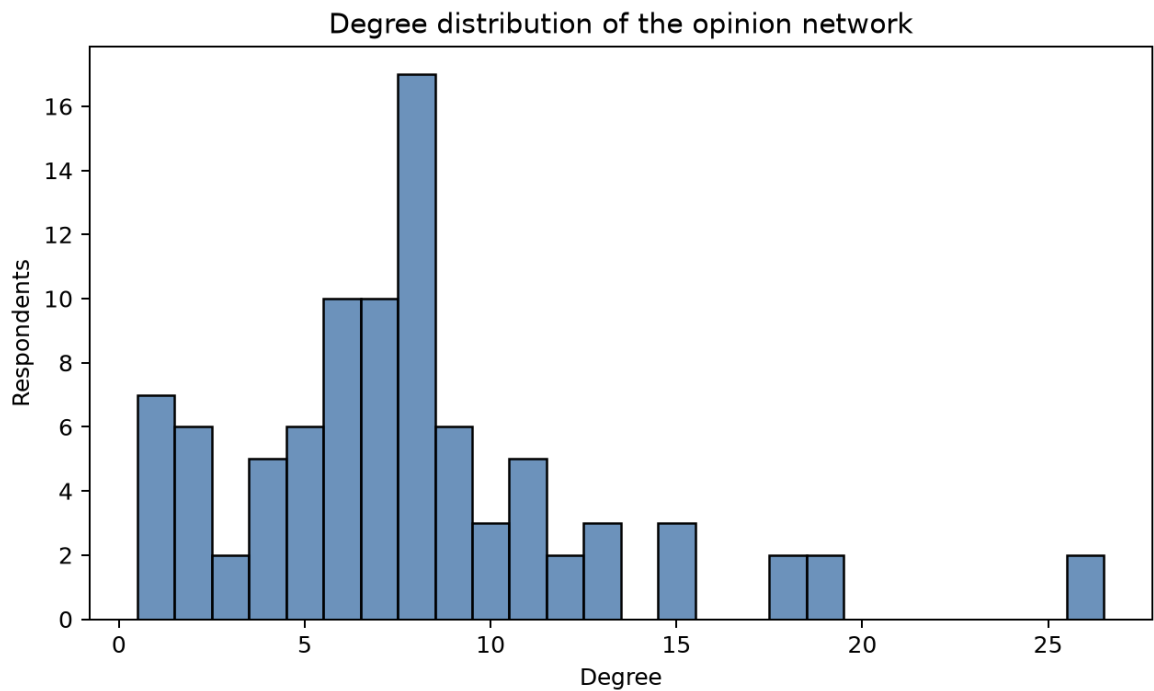


Figure 3. Degree distribution. In a union kNN graph the typical degree is near 8, but hubs can exceed 8 when many students list the same person as a neighbor. A long right tail would be more typical of a scale-free social graph than of this construction.

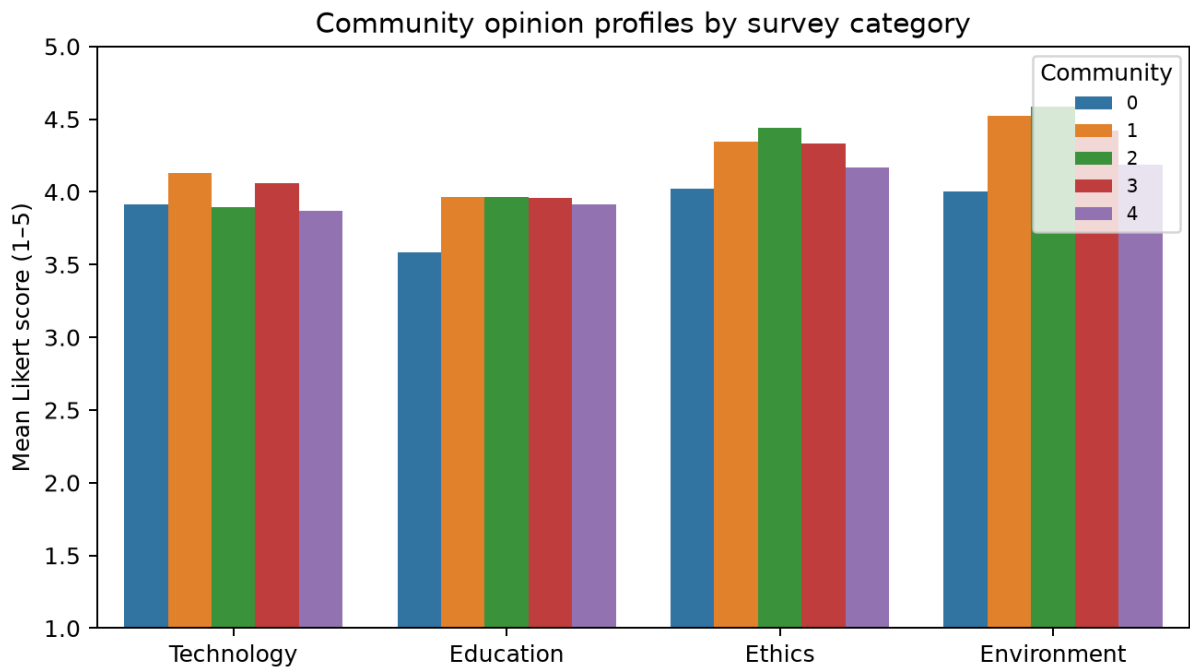


Figure 4. Mean Likert score (1-5) of each community on Technology, Education, Ethics, and Environment. This is the main interpretive plot: communities are not just algorithmic partitions, they have distinct opinion profiles.

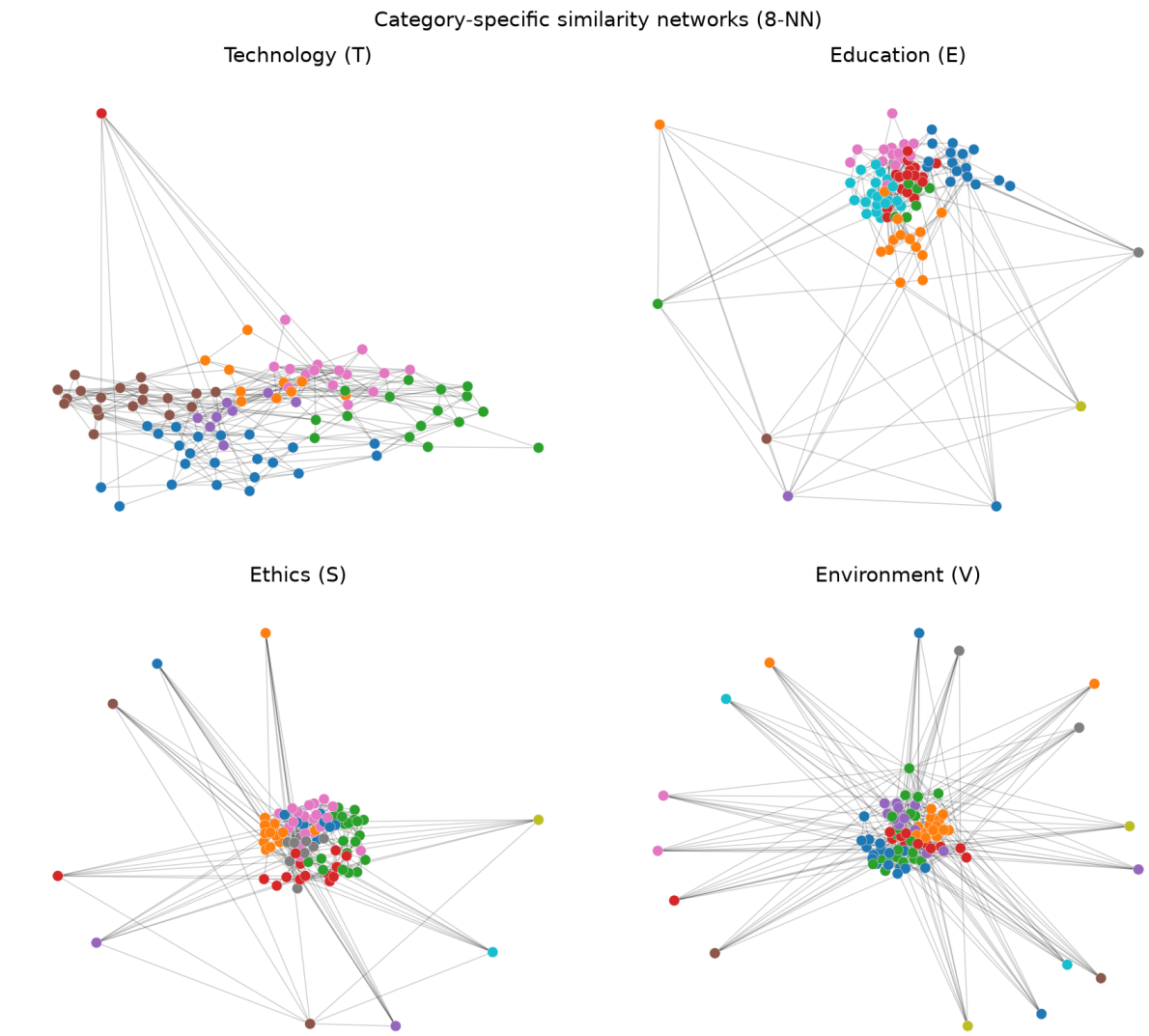


Figure 5. Four category-specific 8-NN networks. Visual fragmentation differs by theme: some themes produce denser, more mixed graphs; others split into clearer blobs. Quantitative comparison is in the modularity bars below.

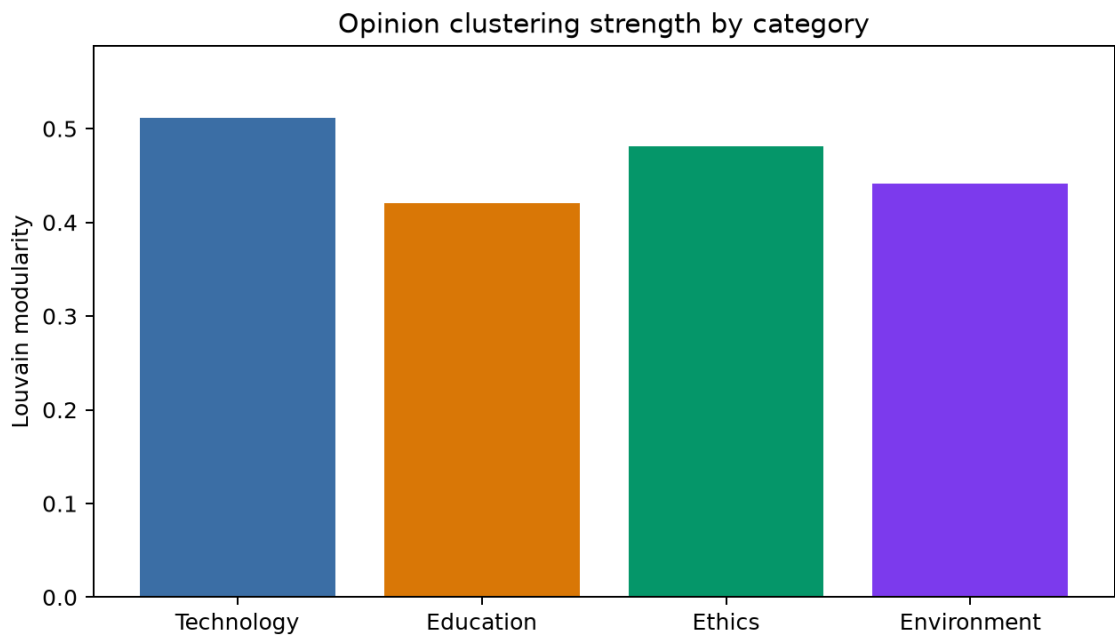


Figure 6. Louvain modularity of each category network. Technology: 354 edges, density 0.086, clustering 0.211, modularity 0.511, 7 communities. Education: 367 edges, density 0.090, clustering 0.206, modularity 0.421, 13 communities. Ethics: 393 edges, density 0.096, clustering 0.172, modularity 0.481, 16 communities. Environment: 432 edges, density 0.105, clustering 0.144, modularity 0.442, 22 communities.

6. Results and Discussion

The class does not hold a single undifferentiated opinion. Mutual nearest-neighbor ties and a Louvain modularity of 0.369 show that respondents cluster into 5 groups. Because items are positively worded, the main axis of variation is how strongly people agree rather than a left-right ideological split. Still, the category means differ enough to tell a story about the cohort.

Community 0 (n=11, overall mean 3.88): Technology=3.91, Education=3.59, Ethics=4.02, Environment=4.00.

Community 1 (n=23, overall mean 4.24): Technology=4.13, Education=3.97, Ethics=4.34, Environment=4.52.

Community 2 (n=15, overall mean 4.22): Technology=3.90, Education=3.97, Ethics=4.44, Environment=4.59.

Community 3 (n=20, overall mean 4.19): Technology=4.06, Education=3.96, Ethics=4.33, Environment=4.42.

Community 4 (n=22, overall mean 4.03): Technology=3.87, Education=3.91, Ethics=4.17, Environment=4.19.

Community 1 is the high-agreement core (overall 4.24): these students tend to endorse AI in education and science, compulsory academic honesty, inclusive campus norms, and environmental action. Community 0 is relatively more reserved (overall 3.88). A lower Technology score often reflects caution about AI in healthcare, autonomous vehicles, or regulation rather than blanket techno-skepticism; a lower Education score can reflect disagreement with compulsory attendance or with replacing exams. The network therefore separates 'enthusiastic endorsers' from 'selective endorsers', which is a realistic picture of an engineering class. Community means still sit in a narrow band (roughly 3.9 to 4.2 overall). Pearson clustering is therefore detecting differences in which items people relatively endorse - for example stronger Environment than Technology, or the reverse - not two hostile camps. That is the right reading of a class survey full of positively worded items.

Across themes, Technology shows the strongest clustering (modularity 0.511), while Education is the most mixed (modularity 0.421). High modularity means students who agree on that theme also tend to agree with a stable set of peers; low modularity means the class is closer to a consensus on that theme. Environment items in this survey are almost uniformly endorsed, so we expect weaker splits there than on Technology (AI, robots, regulation), where the class is more divided.

Centrality adds a second reading. High-eigenvector students sit in the dense agreement core: their answers are typical of a well-connected cluster, so they are useful as illustrative 'median voices' of that cluster, not as influencers in a social sense. High-betweenness students sit between clusters; if this were an actual discussion network, they would be the people most able to translate one group's concerns to another. We cannot claim social influence from this dataset - only structural position in opinion space.

Limitations. (1) Similarity is not interaction: two students with similar answers may never have spoken. (2) kNN sparsification is a modeling choice; a different k would change density and possibly the number of communities, though the similarity heatmap is independent of k. (3) Pearson emphasizes the shape of a profile (which items someone relatively agrees with) rather than overall enthusiasm. (4) Blank cells and 'No Comments' answers are not guaranteed to be random. These caveats do not erase the main finding: the cohort's opinions form a clustered similarity network, with a high-agreement core and smaller, more cautious groups, and some themes fragment the class more than others.

7. Individual Contribution

Fill in real names before submission. The breakdown below matches the actual pipeline so it can be edited rather than rewritten.

- [Member 1] Dataset documentation, Likert encoding, missing-value policy, category tagging.
- [Member 2] Similarity matrix, union kNN graph construction, category sub-networks.
- [Member 3] Global metrics, centrality, Louvain communities, community opinion profiles.
- [Member 4] Visualizations, results write-up, report assembly, README and reproducibility.

If the team has fewer than four members, merge the last rows. All members should review the GitHub repository layout and the final PDF before the 20 September 2026 deadline.

Appendix. Metric definitions

Density is the fraction of possible undirected pairs that are realized as edges. Average degree is $2m/n$. Average clustering is the mean of the weighted local clustering coefficients (NetworkX). Average shortest path and diameter are computed on the largest connected component with hop distance (unweighted). Degree centrality in the tables is the weighted degree (sum of incident Pearson weights). Betweenness is the fraction of shortest paths that pass through a node, with path length 1-similarity so that stronger agreement is a shorter link. Eigenvector centrality scores nodes that are similar to other high-similarity nodes. Louvain modularity compares the weight inside detected communities with the weight expected in a random graph with the same degree sequence. Community profile values are mean Likert scores (1-5) of members of that community, averaged first across items in a category and then across members.

Reproducibility. From the project root, create a virtual environment, install requirements.txt, and run `python run_pipeline.py`. The script writes figures under `outputs/figures/`, numeric tables as CSV/JSON under `outputs/`, and this PDF under `report/report.pdf`. Randomness is fixed by layout and Louvain seeds (42).